

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 11: Credit Score Classification using Machine Learning

Aim

To implement a machine learning program for classifying customers into different credit score categories based on their financial attributes.

Procedure

1. Import the required libraries.
2. Create a sample credit score dataset.
3. Split the dataset into training and testing sets.
4. Train a Random Forest classification model.
5. Predict the credit score category of test data.
6. Calculate the classification accuracy.
7. Classify a new customer.
8. Display the result.

Algorithm

1. Start.
2. Load the credit score dataset.
3. Separate input features and target classes.
4. Split the dataset into training and testing data.
5. Train the classification model.
6. Predict credit score classes for test data.
7. Evaluate the model using accuracy.
8. Predict the credit score category for a new customer.
9. Stop.

Dataset 1 — Customer Financial Profile

Age	Annual_Income	Credit_History	Loan_Amount	Credit_Score
22	25000	1	10000	Poor
28	35000	1	12000	Fair
35	50000	2	15000	Good
42	65000	3	18000	Excellent
50	80000	4	20000	Excellent

Dataset 2 — Loan Repayment Profile

Monthly_Income	Existing_Loan	Repayment_History	Credit_Utilization	Credit_Score
18000	8000	2	85	Poor
25000	7000	3	70	Fair
35000	5000	4	55	Good
45000	4000	5	40	Good
60000	2000	6	25	Excellent

Dataset 3 — Credit Card Profile

Age	Monthly_Spending	Payment_Delay	Credit_Limit	Credit_Score
21	18000	6	30000	Poor
26	15000	4	35000	Fair
32	12000	2	50000	Good
38	10000	1	60000	Good
45	8000	0	80000	Excellent

Dataset 4 — Employment and Credit Profile

Age	Years_Employed	Annual_Income	Loan_Defaults	Credit_Score
23	1	22000	3	Poor
27	2	30000	2	Fair
34	5	45000	1	Good
40	8	55000	0	Good
48	12	70000	0	Excellent

Dataset 5 — Banking Transaction Profile

Monthly_Income	Savings	Transactions	Overdrafts	Credit_Score
20000	3000	25	5	Poor
28000	6000	30	4	Fair
40000	12000	35	2	Good
50000	20000	40	1	Good
65000	35000	45	0	Excellent

Code

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

X = [
    [25, 25000, 1, 2],
    [30, 40000, 3, 1],
    [35, 60000, 5, 0],
    [45, 80000, 8, 0],
    [50, 90000, 10, 0],
    [22, 20000, 0, 4],
    [28, 30000, 2, 3],
    [40, 70000, 7, 1],
    [55, 100000, 12, 0],
    [24, 22000, 1, 4],
    [32, 50000, 4, 1],
    [48, 85000, 9, 0],
    [27, 28000, 1, 3],
    [38, 65000, 6, 1],
    [52, 95000, 11, 0]
]

y = [
    "Poor", "Average", "Good", "Good", "Good",
    "Poor", "Poor", "Good", "Good", "Poor",
    "Average", "Good", "Poor", "Good", "Good"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

new_customer = [[35, 55000, 5, 1]]
prediction = model.predict(new_customer)
print("Credit Score Classification:", prediction[0])
```

Output

```
*** Actual Labels: ['Poor', 'Good', 'Poor', 'Good', 'Poor']
Predicted Labels: ['Poor' 'Good' 'Poor' 'Good' 'Poor']
Accuracy: 1.0
Confusion Matrix:
[[2 0]
 [0 3]]
Credit Score Classification: Average
```

Result

The Credit Score Classification program was successfully implemented using a Random Forest classifier. The model classified customers into Poor, Average, and Good credit score categories and predicted the category of a new customer.